Homework 2

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1. Question 1

1.a.

Yes, see reasoning in 1b.

1.b.

Define the payoff as X, price vector as P, and price of contingent claim as p. Solve the linear equation $Xp = P$ by elimination, we got

$$[X, P] \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0.2 \\ 0 & 1 & 0 & 0.4 \\ 0 & 0 & 1 & 0.2 \end{bmatrix} \tag{1}$$

Thus, we have complete markets in this economy and the price of continguent claim for state of nature 1, 2,3 are $[0.2, 0.4, 0.2]'$

1.c.

The price should be

$$[30, 20, 10] \times [0.2, 0.4, 0.2]' = 16 \tag{2}$$

1.d.

SDP in this economy is calucalteed by $M(i) = \frac{p(i)}{\pi(i)}$

$$M = [0.8, 0.8, 0.8]'. \tag{3}$$

This economy satisfies no-arbitrage as all the prices of contingent claims are strictly positive and unique. SDF remains the same accross all states, which means that agents in this economy are risk neutral.

1.e.

$$P_D = \sum_i \pi(i) M(i) X(i) = \frac{1}{4} \cdot 0.8 \cdot 30 + \frac{1}{2} \cdot 0.8 \cdot 20 + \frac{1}{4} \cdot 0.8 \cdot 10 = 16 \tag{4}$$

Homework
Solution

Published Oct 03, 2026

2. Question 2

2.a.

Parameter	Estimate	Std. error	t-stat
a	1.0368	0.0119	86.86
b_1 (Lo 10)	−1.5850	1.6173	−0.98
b_2 (2-Dec)	3.5503	1.8540	1.91
b_3 (3-Dec)	0.7104	2.0766	0.34
b_4 (4-Dec)	−3.1122	1.8756	−1.66
b_5 (5-Dec)	1.1837	2.0043	0.59
b_6 (6-Dec)	2.9972	1.9857	1.51
b_7 (7-Dec)	−5.2931	1.8685	−2.83
b_8 (8-Dec)	2.4722	1.6310	1.52
b_9 (9-Dec)	2.8775	1.6288	1.77
b_{10} (Hi 10)	−1.1771	1.0598	−1.11

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 19.

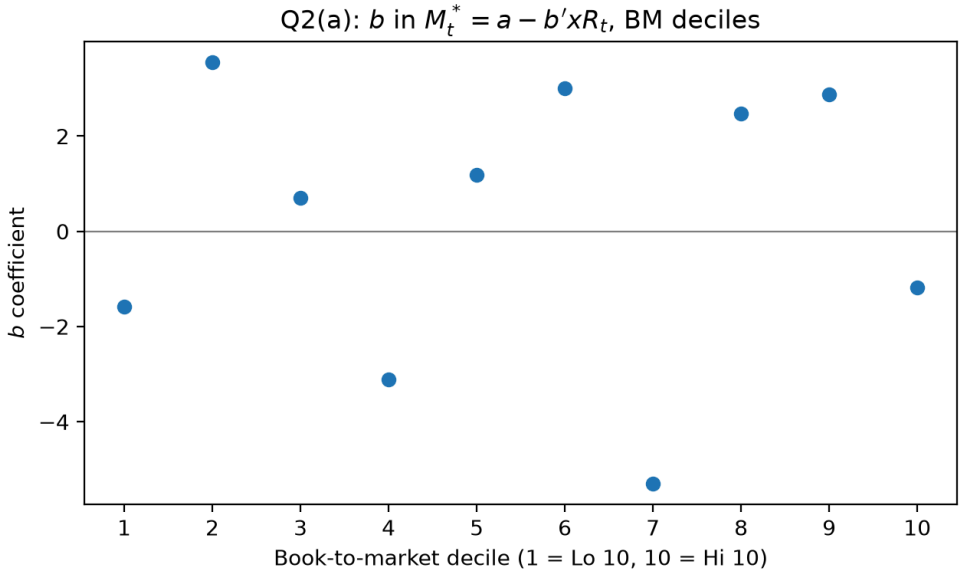


Figure 1: b coefficients of $M_t^* = a - b'xR_t$ by book-to-market decile (two-step GMM, 1927:07–2026:08).

2.b.

Parameter	Estimate	Std. error	t-stat
a	1.0666	0.0154	69.32
b_1 (Lo PRIOR)	−4.0194	0.9747	−4.12

Parameter	Estimate	Std. error	t-stat
b_2 (PRIOR 2)	1.1429	1.4347	0.80
b_3 (PRIOR 3)	0.1024	1.8005	0.06
b_4 (PRIOR 4)	1.4897	1.7155	0.87
b_5 (PRIOR 5)	2.5317	2.0073	1.26
b_6 (PRIOR 6)	−0.8195	1.8827	−0.44
b_7 (PRIOR 7)	0.2777	1.8301	0.15
b_8 (PRIOR 8)	0.9818	1.7812	0.55
b_9 (PRIOR 9)	−0.9414	1.6701	−0.56
b_{10} (Hi PRIOR)	3.4370	0.8792	3.91

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 3.

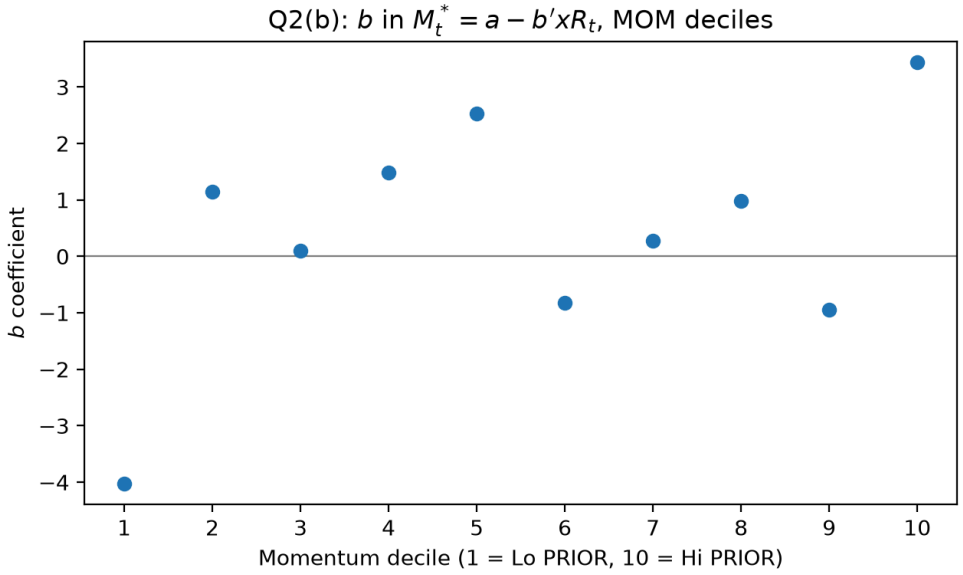


Figure 2: b coefficients of $M_t^* = a - b'xR_t$ by momentum decile (two-step GMM, 1927:07–2026:08).

The sample correlation between M_t^* from Question 2(a) (BM deciles) and M_t^* from Question 2(b) (MOM deciles) is $\mathrm{Corr}\left(M_{BM,t}^*, M_{MOM,t}^*\right) = 0.3555$ (monthly, 1927:07–2026:08, $T = 1190$).

2.c.

Parameter	Estimate	Std. error	t-stat
a	1.0927	0.0195	56.14
b_1 (Lo 10)	−1.9664	1.8048	−1.09
b_2 (2-Dec)	2.4739	1.8751	1.32
b_3 (3-Dec)	0.5781	2.1303	0.27
b_4 (4-Dec)	−3.7439	1.8387	−2.04

Parameter	Estimate	Std. error	t-stat
b_5 (5-Dec)	1.0655	2.1028	0.51
b_6 (6-Dec)	2.6840	1.7591	1.53
b_7 (7-Dec)	−4.6456	1.7091	−2.72
b_8 (8-Dec)	2.4707	1.6922	1.46
b_9 (9-Dec)	3.7398	1.6604	2.25
b_{10} (Hi 10)	−0.2415	1.0046	−0.24
b_{11} (Lo PRIOR)	−4.4169	1.1081	−3.99
b_{12} (PRIOR 2)	0.6354	1.5542	0.41
b_{13} (PRIOR 3)	−0.1712	1.8918	−0.09
b_{14} (PRIOR 4)	1.9480	1.8068	1.08
b_{15} (PRIOR 5)	1.4236	2.1083	0.68
b_{16} (PRIOR 6)	−0.8781	2.2007	−0.40
b_{17} (PRIOR 7)	−0.1639	1.8900	−0.09
b_{18} (PRIOR 8)	0.9802	1.8749	0.52
b_{19} (PRIOR 9)	−1.2708	1.8496	−0.69
b_{20} (Hi PRIOR)	3.4473	1.0952	3.15

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 7.

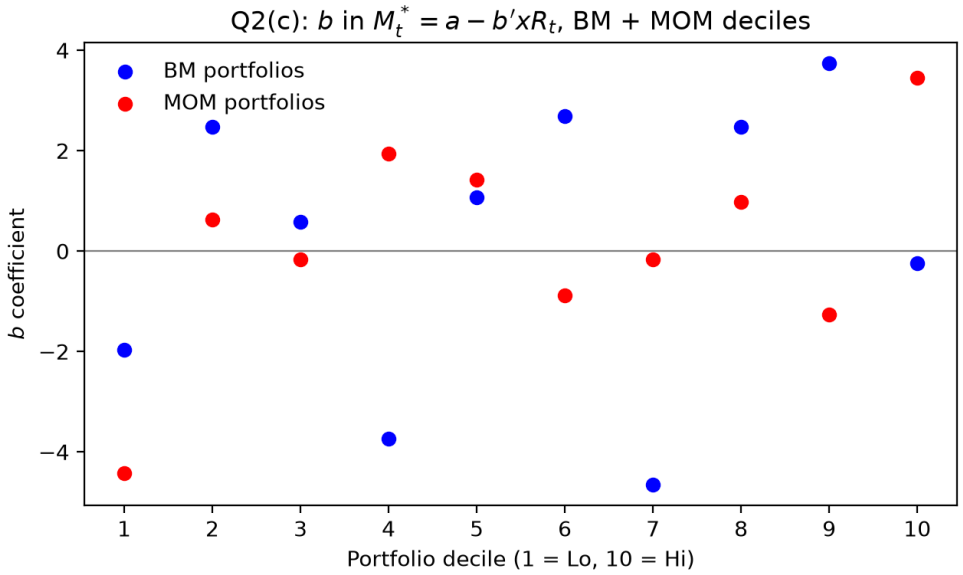


Figure 3: b coefficients of $M_t^* = a - b'xR_t$ estimated on the 20 BM and MOM decile portfolios jointly, by decile (blue: BM portfolios; red: MOM portfolios; two-step GMM, 1927:07–2026:08).

Question	Test assets	Annualized Sharpe ratio of $xR_t^* = b'xR_t$

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2(a)	10 BM deciles	0.6646
2(b)	10 MOM deciles	0.8937
2(c)	20 BM + MOM deciles	1.0545

Monthly 1927:07–2026:08 (T = 1190); Sharpe ratio = $\sqrt{12}$ × mean / s.d. of the monthly xR_t^* series.

3. Question 3

3.a.